9.6.2 Enhanced control and data transmission for long-range and complex drone missions

Advanced communication networks, such as 5G, play a crucial role in enabling drones to undertake long-range, multi-phase, and high-complexity missions that require uninterrupted command links and continuous data exchange. Traditional radio and Wi-Fi systems provide only limited range and often suffer from latency, signal interference, and reduced reliability when drones operate beyond visual line of sight. In contrast, 5G networks offer high-capacity data channels and real-time responsiveness, allowing UAVs to receive flight instructions, environmental updates, and mission adjustments instantaneously during operation. This ensures that drones can navigate accurately through changing conditions, avoid obstacles, and respond to mission-critical updates without manual intervention delays.

High-speed data transmission is particularly essential when drones are equipped with high-resolution cameras, LiDAR scanners, thermal sensors, or multispectral instruments, as these systems generate large volumes of data that must be processed and transmitted quickly for real-time decision-making. Through 5G and future hybrid networks that combine cellular, satellite, and edge computing, drones can stream data directly into cloud-based analytical platforms, enabling ground teams or automated AI systems to interpret results and update mission parameters immediately. This capability is vital in infrastructure inspection, precision agriculture, maritime surveillance, cross-border logistics, and emergency relief operations, where situational awareness and fast information feedback determine mission success.

By supporting both command-and-control stability and high-throughput data exchange, advanced communication networks transform drones into intelligent, fully networked aerial assets capable of performing coordinated